TradePilot v4 -- Complete Glossary of Trading & ML Terms

A plain-English reference for every concept used in the TradePilot platform. Written for smart people who are new to technical trading -- no jargon to explain jargon.

Section 1: Market & Trading Basics

Term	What It Is	Real-World Analogy	How TradePilot Uses It
Nifty 50	An index of the 50 largest companies listed on NSE (National Stock Exchange). It represents the overall health of the Indian stock market.	Think of it as the "temperature" of the Indian economy -- if Nifty is up, most big companies are doing well.	TradePilot scans all 50 Nifty stocks every day to find the best ones to buy. Our entire universe is these 50 stocks.
NSE	National Stock Exchange of India -- the primary stock exchange where Nifty 50 stocks trade. Based in Mumbai.	Like a supermarket for stocks -- buyers and sellers meet here to trade.	All our data comes from NSE -- prices, volumes, options data, FII/DII flows.
BSE	Bombay Stock Exchange -- the older Indian exchange. Same stocks trade on both NSE and BSE.	Like having two supermarkets in the same city selling the same products at nearly the same prices.	We primarily use NSE data (`.NS` suffix in code), but some stocks also have `.BO` (BSE) data.
Intraday Trading	Buying and selling a stock on the same day. You start the day with cash, buy stocks, sell them before market close (3:30 PM), and end the day with cash again.	Like buying vegetables in the morning market and selling them the same evening -- you don't hold inventory overnight.	TradePilot v4 is designed for intraday signals. We scan at 9:30 AM and suggest stocks to buy and sell within the same day.
Swing Trading	Holding stocks for 2-10 days. You buy today	Like buying a product on sale and waiting a	v3 predicted 5-day returns (swing trading). v4 shifted to intraday

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	and sell after a few days when the price moves in your favour.	week for the price to go back up before reselling.	because our users want same-day results.
Bull Market	When the market is going up consistently. Prices rise, optimism is high.	Like a rising tide that lifts all boats -- most stocks go up in a bull market.	TradePilot detects "BULL" regime when Nifty is up more than +0.5% for the day. In bull markets, we are more aggressive with BUY signals.
Bear Market	When the market is falling consistently. Prices drop, fear is high.	Like a winter sale that never ends -- everything keeps getting cheaper.	Detected as "BEAR" when Nifty is down more than -0.5%. We become cautious and reduce BUY signals.
Sideways Market	Market is flat -- not clearly going up or down. Prices bounce in a narrow range.	Like a car stuck in traffic -- moving a little forward, a little back, but not really going anywhere.	Labelled "NEUTRAL" in our code. Nifty change is between -0.5% and +0.5%.
Market Regime	The current "mood" of the market -- bull, bear, or sideways. It changes the rules of what works.	Like weather -- you dress differently for rain vs sunshine. Trading strategies that work in bull markets can fail in bear markets.	v4 detects regime from Nifty's daily change and adjusts scoring. In v3, different BUY thresholds were used for different regimes (50-60 range).
Stop-Loss (SL)	A pre-decided price at which you exit a losing trade to limit your loss. If you buy at Rs 100 with a 2% SL, you sell at Rs 98.	Like setting an alarm -- "If the temperature drops below 15 degrees, I'm wearing a jacket." You decide in advance when to cut your losses.	TradePilot calculates SL% for every stock based on its score and volatility. Higher-score stocks get tighter SL (1%), lower-score stocks get wider SL (2%).
Target	The price at which you plan to sell for a profit. If you buy at Rs 100 with a 2.5% target, you sell at Rs 102.50.	Like setting a goal -- "I'll sell my second-hand phone when someone offers Rs 15,000." You know when to take your profit.	Computed per stock. High-score stocks get wider targets (2.5%) because we're more confident. Calculated in `composite_scorer.py`.
Trailing Stop	A stop-loss that moves up as the price goes up. If the stock rises from 100 to 110, your SL moves from 98 to 108. It locks in profits.	Like a ratchet -- it only moves in one direction. Your safety net keeps rising as the stock climbs.	Referenced in our strategy research. Not yet implemented in v4 but planned for the paper trading pool.

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Paper Trading	Simulated trading with fake money to test strategies without risking real capital. Everything works like real trading except no actual money changes hands.	Like a flight simulator -- pilots practice before flying real planes. We test our algorithm before putting real money on the line.	TradePilot runs a paper trading pool (Rs 10,00,000 simulated capital) to track how our BUY signals would have performed.
Position	A single stock holding in your portfolio. If you bought 10 shares of Reliance, that's one position.	Like one item in your shopping cart.	Each BUY signal becomes a position with a specific quantity, entry price, SL, and target.
Portfolio	All your positions combined. If you hold Reliance, TCS, and HDFC Bank -- that's your portfolio.	Your entire collection of investments -- like all the items in your shopping cart together.	TradePilot manages a portfolio of 5-10 BUY positions per day, tracked in `position_sizer.py`.
Capital Deployment	How much of your total money you've actually invested in stocks (vs keeping as cash).	If you have Rs 10 lakh but only invested Rs 7 lakh, your deployment is 70%. The remaining 30% is cash "on the bench."	v4 tracks `total_deployed_pct` -- typically 70-90% of capital is deployed across BUY signals.
Risk-Reward Ratio (R:R)	How much you could gain vs how much you could lose. A 1:2 ratio means for every Rs 1 you risk, you could gain Rs 2.	Like a bet -- "I'll risk Rs 100 to potentially win Rs 200." Good bets have R:R of 1:2 or better.	Calculated for every position: `riskReward = target / stopLoss`. We aim for R:R above 1.5.

Section 2: Technical Indicators (What Our Algorithm Uses)

Term	What It Is	Real-World Analogy	How TradePilot Uses It	Example (Apr 8, 2026)
RSI (Relative Strength Index)	A 0-100 score measuring if a stock has gone up "too much" (overbought, >70) or down "too much" (oversold,	Like a thermometer for stocks. Above 70 means "fever" (overheated,	One of our Tier 1 features. RSI(14) was the top SHAP feature in ML research. v4 uses it as a mean-reversion signal.	RSI 50 = neutral. If TITAN's RSI was 72 after its 3.97% rally, it

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	<30). Uses last 14 days of price data.	may cool down). Below 30 means "hypothermia" (oversold, may bounce back).		might be overheated.
MACD (Moving Average Convergence Divergence)	Shows the relationship between two moving averages (12-day and 26-day). When the fast line crosses above the slow line, it's a buy signal.	Like comparing your recent exam score to your yearly average. If recent scores are rising faster than your average, you're on an upward trend.	Tier 2 feature. Was 7.56% feature importance in v3. v4 uses MACD histogram for trend strength estimation.	If MACD says "Bullish" for RELIANCE, it means recent momentum is accelerating upward.
MACD Histogram	The visual difference between the MACD line and its signal line. Positive bars = bullish momentum growing. Negative bars = bearish.	Like the speedometer needle -- not just "are you moving?" but "are you accelerating or slowing down?"	Used as a Tier 2 feature in v4. Positive histogram + positive price change = "Bullish" MACD label.	A growing positive histogram means the bullish trend is strengthening.
SMA (Simple Moving Average)	The average closing price over N days. SMA(20) = average of last 20 days. Smooths out daily noise to show the trend.	Like your running average marks in school -- one bad test doesn't change it much, but a string of bad tests pulls it down.	Used in Golden Cross / Death Cross detection. Also used to calculate volume ratio (today's volume vs 20-day average).	If Nifty's 50-day SMA is 22,100 and current level is 22,400 -- market is above its average (bullish).
EMA (Exponential Moving Average)	Like SMA but gives MORE weight to recent prices. Reacts faster to new information. EMA(12) moves quicker than SMA(12).	Like a weighted GPA where your most recent semester counts more than older ones.	MACD is built using EMAs (12-day and 26-day). EMA reacts faster to price changes than SMA.	EMA catches trend changes 1-2 days before SMA does.

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Bollinger Bands	Three lines around price: middle = 20-day SMA, upper band = SMA + 2 standard deviations, lower band = SMA - 2 standard deviations. Price usually stays within the bands.	Like a highway with guardrails. When the car (price) hits the guardrail, it usually bounces back toward the centre. Occasionally it crashes through (breakout).	Used via Bollinger %B in Tier 2 features. Helps identify when a stock is at extreme levels.	If SBIN price touches the lower Bollinger Band, it might be oversold and due for a bounce.
Bollinger %B	A 0-1 score showing where price sits within Bollinger Bands. 0 = at lower band, 0.5 = at middle, 1 = at upper band. Above 1 or below 0 = outside bands.	Like a floor indicator in an elevator. 0 = ground floor, 0.5 = middle floor, 1 = top floor. Going above 1 = you're on the roof.	Tier 2 feature. Values near 0 suggest mean-reversion buy opportunity. Values near 1 suggest the stock may be overextended.	%B of 0.15 for INDUSINDBK means it's near the lower band -- potential bounce candidate.
ATR (Average True Range)	Measures how much a stock typically moves in a day (in rupees). A stock with ATR of Rs 50 moves about Rs 50 per day on average.	Like knowing a child's energy level -- some kids run around a lot (high ATR), others sit quietly (low ATR). ATR tells you how "wild" a stock is.	Was the TOP feature in v3 (13.25% importance). v4 uses ATR(14)/close as normalized volatility. Used to set stop-loss distances.	If TITAN has ATR of Rs 45 and price is Rs 3200, normalized ATR = 1.4%. This helps set a proportional SL.
ADX (Average Directional Index)	Measures trend strength on a 0-100 scale. Above 25 = strong trend. Below 20 = no clear trend (ranging/sideways). Does NOT tell direction, just strength.	Like a wind speed meter -- it tells you how strong the wind is, but not which direction it's blowing.	Tier 2 feature. ADX(14) helps distinguish trending markets (where momentum works) from ranging markets (where mean-reversion works).	ADX of 30 for BAJFINANCE means it's in a strong trend. ADX of 12 means it's going sideways.
OBV (On Balance Volume)	A running total that adds volume on up-days and subtracts	Like counting votes -- if more people "vote"	Referenced in research as a top feature. Volume-based	If MARUTI's price is flat but OBV is rising,

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	volume on down-days. Rising OBV = more volume behind upward moves.	(trade volume) on up days than down days, the stock has popular support for going higher.	confirmation of price trends.	smart money may be quietly accumulating.
Volume Ratio	Today's trading volume divided by the 20-day average volume. Ratio of 2.0 means twice the normal volume.	Like footfall at a shop -- if usually 100 customers come per day but today 200 showed up, something special is happening.	Key v4 feature (`stock_volume_ratio`). Volume surge often precedes big price moves. Ratio > 1.5 is flagged as "Volume surge."	SHRIRAMFIN with volume ratio 2.3x means 130% more trading than usual -- something is driving interest.
Golden Cross	When the 50-day SMA crosses ABOVE the 200-day SMA. Considered a major bullish signal.	Like a student whose recent performance just overtook their long-term average -- they're improving and gaining momentum.	Referenced in research. Signals potential start of a long-term uptrend.	If RELIANCE's 50-day average crosses above its 200-day average, traders see it as a strong buy signal.
Death Cross	Opposite of Golden Cross. The 50-day SMA crosses BELOW the 200-day SMA. Bearish signal.	Recent performance dropping below long-term average -- like a company whose quarterly results keep getting worse than their annual trend.	Referenced in research. Signals potential start of a long-term downtrend.	A Death Cross on HDFC Bank would cause many traders to sell or avoid the stock.
52-Week High/Low	The highest and lowest price a stock reached in the past 1 year. Stocks near 52-week high are often in strong uptrends.	Like checking if your salary is at its highest-ever level. Being near your all-time high suggests strong	v3 used `pct_from_high` and `pct_from_low` as features (6.21% importance). Stocks near 52-week high have momentum.	If TITAN at Rs 3200 has a 52-week high of Rs 3250, it's 1.5% away -- very close, suggesting

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		career momentum.		strong momentum.

Section 3: Intraday Signals (v4 New Features)

These are the features that make v4 different from v3. They focus on what happens DURING the trading day.

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VWAP (Volume Weighted Average Price)	The average price of a stock weighted by volume. If more shares traded at Rs 100 than at Rs 95, VWAP will be closer to Rs 100. Calculated from market open.	Like the "true average" price everyone paid today. If 90% of customers bought a shirt at Rs 500 and 10% at Rs 400, the VWAP is closer to Rs 500, not the simple average of Rs 450.	The #1 institutional indicator. v4 computes VWAP from intraday OHLCV candles. Price above VWAP = bullish (+score), below = bearish (-score). Gets 10% weight in composite score.
ORB (Opening Range Breakout)	The high and low of the first 15 minutes of trading (9:15-9:30 AM). If the stock breaks above this high, it's a bullish breakout. Below the low = bearish breakdown.	Like the first 15 minutes of a cricket match -- if the opening batsmen score fast, the team usually has a strong innings. The early pace sets the tone.	Core v4 signal (15% weight in composite). Computed in `features_intraday.py`. Breakout direction (+1, -1, or 0) and strength are calculated. Research shows ORB has 60-89% win rate on NSE.
Gap Up / Gap Down	When a stock opens at a significantly different price than yesterday's close. Gap Up = opens higher. Gap Down = opens lower.	Like waking up to find your house value jumped overnight due to a new metro station announcement. The price "jumped" without any trading in between.	v4 computes gap% = (open - prev_close) / prev_close. Classified as small (<0.5%), medium (0.5-1.5%), or large (>1.5%). Gaps are predictive of the day's direction.
Gap Analysis	Classifying the gap's size and likely behaviour. Large gaps often partially	Like a rubber band -- stretch it a lot (large gap) and it snaps back. Stretch it a little	Computed in `compute_gap_analysis()`. Gap type (up/down/flat) and magnitude

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	fill (gap fill). Small gaps often continue in the gap direction.	(small gap) and it stays.	(small/medium/large) feed into the momentum score.
First Hour Return	How much a stock moved in the first hour of trading (9:15-10:15 AM). A positive first hour often predicts a positive day.	Like the first chapter of a book -- if it hooks you, the whole book is usually good. A strong start often leads to a strong finish.	Computed in `compute_intraday_momentum()`. First hour return is a momentum indicator that captures early institutional activity.
Volume Acceleration	How the first 30 minutes of trading volume compares to the average volume per period. High acceleration = institutions are active early.	Like a restaurant that's packed at opening time vs one that fills up slowly. Early rush suggests high demand (maybe a positive review just came out).	Computed as first_30_min_vol / avg_candle_vol. Values above 1.5 mean "Volume surge (1.5x average)" -- a positive signal.
Relative Strength (vs Market)	How much a stock outperforms or underperforms the Nifty 50 index TODAY. If Nifty is +0.5% and a stock is +2%, its RS = +1.5%.	Like comparing your exam marks to the class average. Scoring 85% when the class average is 70% means you have strong relative performance (+15 points).	Core v4 signal (20% weight -- the highest!). Stocks are ranked by RS among all 50 Nifty stocks. Top performers get higher scores. This is the #1 fix from v3's diagnosis.
Relative Strength Ranking	After computing RS for all 50 stocks, ranking them from best to worst. The top-ranked stock gets a percentile score near 1.0, the bottom gets near 0.0.	Like class rank -- not just your marks, but your position in the class. Being #3 out of 50 is better than being #25, even if both scored well.	Percentile ranking is used directly in composite scoring. `rs_score = rank / total_stocks`. This ensures we always pick the STRONGEST stocks, not just "good enough" ones.

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Intraday Range	The difference between the day's highest and lowest price, as a percentage. High range = volatile day. Low range = calm day.	Like measuring how much temperature varied during the day. A range of 5-35 degrees (high) vs 22-28 degrees (low).	Computed as `(day_high - day_low) / day_low * 100`. Used to estimate volatility: >3% = High, >1.5% = Medium, <1.5% = Low. Affects SL/target sizing.
Pre-market Session	A 15-minute window (9:00-9:15 AM) before the market opens where orders are collected but not executed. The "equilibrium price" discovered here becomes the opening price.	Like an auction preview -- people submit their bids before the auction starts. The auctioneer figures out the starting price based on all the bids.	v4 architecture plans to use pre-market data (via nsefin library) at 9:00 AM as Layer 1 -- before the market even opens. This gives a head start on the day's direction.

Section 4: Institutional & Options Data

This is the "smart money" layer -- what the big players (FIIs, DIIs) and the options market are telling us.

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FII (Foreign Institutional Investors)	Foreign funds (like Goldman Sachs, BlackRock) that invest in Indian stocks. They control a huge chunk of Nifty 50 ownership.	Like a whale in a swimming pool -- when FIIs move, the whole market feels it. Their buying or selling creates waves.	v4 tracks FII net flow daily (in crores). FII buying = bullish signal. FII data gets 10% weight in composite score.
DII (Domestic Institutional Investors)	Indian funds like LIC, SBI Mutual Fund, HDFC AMC. They often buy when FIIs sell (providing "support").	Like local businessmen in a town -- when outsiders leave, locals keep things running. DIIs often stabilize the market during FII selling.	Tracked alongside FII. Both FII+DII buying = strongest bullish signal (score +1.0). DII buying alone = weaker signal (+0.1) because DIIs are often reactive.

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FII/DII Net Flow	The difference between buying and selling by FIIs or DIIs. Net positive = they bought more than they sold. Net negative = they sold more. Measured in crores.	Like tracking whether a store had more customers coming in or going out. Net positive footfall = the store is popular today.	Combined into `compute_fii_dii_score()` which produces a -1 to +1 score. Both buying = +1.0 (strong bull). Both selling = -1.0 (bear). The magnitude (size of flow) also matters.
Options	Contracts that give you the RIGHT (but not obligation) to buy or sell a stock at a specific price by a specific date. They're like insurance policies on stocks.	Like booking a hotel room with free cancellation. You pay a small fee (premium) to lock in the price. If you don't want it later, you just lose the booking fee.	TradePilot reads options data to understand what "smart money" expects. Options data is often a LEADING indicator -- it moves before the stock price.
Put Option	The right to SELL a stock at a specific price. You buy puts when you think the stock will go DOWN. More puts being bought = bearish sentiment.	Like buying insurance on your car. You pay a premium, and if something bad happens (stock falls), you get compensated.	Put volume and Open Interest feed into PCR calculation. High put buying can mean fear (bearish) or hedging (neutral).
Call Option	The right to BUY a stock at a specific price. You buy calls when you think the stock will go UP. More calls being bought = bullish sentiment.	Like booking a flat at today's price before the price rises. If prices go up, you win. If they don't, you lose only the booking amount.	Call OI and volume are the other half of PCR. Heavy call buying at higher strikes = bullish expectations.
Open Interest (OI)	The total number of active (unsettled) options contracts. Rising OI = new money entering. Falling OI = positions being closed.	Like the total number of active hotel bookings. Rising bookings = more people committing. Falling bookings = people cancelling.	Core to our OI buildup detection. OI change + price direction = 4 possible patterns (long buildup, short buildup, short covering, long unwinding). Gets 10% composite weight.
PCR (Put-Call Ratio)	Total Put OI divided by total Call OI. PCR > 1.2 = more puts than calls (contrarian	Like a sentiment survey: "Are more people buying umbrellas (puts/insurance) or	Computed from options chain. PCR > 1.2 = "bullish" signal, PCR < 0.7 = "bearish". Used in `compute_oi_buildup()`.

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	bullish). PCR < 0.7 = more calls (contrarian bearish).	sunglasses (calls/optimism)?" If everyone is buying umbrellas, the storm might be over (contrarian).	
Long Buildup	Price UP + OI UP. New buyers are entering the market with conviction. The strongest bullish signal from options data.	Like a restaurant getting busier (more bookings) AND raising prices -- demand is genuinely strong.	OI score = +1.0 (maximum bullish). TradePilot flags this as "Long buildup in options" (positive reason).
Short Buildup	Price DOWN + OI UP. New sellers (shorts) are entering the market. Bearish signal.	Like a restaurant getting more cancellations (short interest) while prices drop -- people are betting against the business.	OI score = -1.0 (maximum bearish). Flagged as "Short buildup in options" (negative reason).
Short Covering	Price UP + OI DOWN. Previous short-sellers are buying back (covering) their positions. Moderately bullish.	Like people who bet against a sports team now scrambling to change their bets because the team is winning.	OI score = +0.3. Flagged as "Short covering rally" (positive reason). The rally may be temporary since it's driven by covering, not new buying.
Long Unwinding	Price DOWN + OI DOWN. Previous buyers are exiting. Moderately bearish -- selling pressure from disappointed longs.	Like early investors in a startup selling their shares because they've lost confidence.	OI score = -0.3. The weakest bearish signal -- existing positions are closing, not new shorts entering.
Max Pain	The price at which the maximum number of options contracts expire worthless. The stock often gravitates toward max pain near expiry.	Like a casino where the house always wins -- max pain is the price where options writers (who are the "house") lose the least money.	v4 computes `max_pain_distance_pct` = how far the current price is from max pain. If price is far from max pain, it may gravitate back.
F&O (Futures & Options)	The derivatives segment of NSE where futures and options contracts trade. Not all stocks	Like the VIP section of the stock market -- only the biggest, most liquid stocks get in. F&O lets	All Nifty 50 stocks are in F&O, so we can get options data for our entire universe. Days to F&O expiry is a Tier 3 feature.

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	have F&O -- only about 200 liquid ones. All Nifty 50 stocks have F&O.	you trade with leverage and hedging.	
Expiry	The date when an options/futures contract expires. Weekly (every Thursday) and monthly (last Thursday). Markets are often volatile on expiry days.	Like the deadline for a sale -- as the deadline approaches, people rush to act. Expiry creates urgency and volatility.	Days to expiry is a Tier 3 feature. Expiry weeks show different trading patterns.
Options Chain	A table showing all available options for a stock -- every strike price, with call and put prices, volumes, and OI.	Like a menu at a restaurant -- all the different "meals" (strike prices) you can order, with their prices and popularity.	TradePilot fetches the full options chain via NSE API to compute PCR, max pain, and OI buildup signals.
Delivery Percentage	The percentage of traded volume that resulted in actual delivery (shares changing hands) vs speculative (intraday) trading. High delivery % = genuine interest.	Like the difference between window shoppers and actual buyers. If 40% of people who entered the shop actually bought something, that's a good conversion rate.	Tier 3 feature. High delivery % (>50%) = conviction buying. Low delivery % (<20%) = mostly speculative trading.

Section 5: Machine Learning Concepts

How TradePilot's "brain" works -- the AI/ML engine that processes all these signals.

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ML (Machine Learning)	Teaching computers to find patterns in data without being explicitly programmed. Give it 3 years of stock data, and it learns which	Like teaching a child by showing examples. Show them 1000 photos of cats and dogs, and they learn to tell them apart without you explaining "cats have whiskers."	TradePilot's core engine. v4 uses ML as one of 7 signals in the composite score (25% weight). ML predicts expected intraday return.

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	patterns lead to stocks going up.		
LightGBM	A fast, efficient ML algorithm that builds hundreds of small decision trees and combines them. Made by Microsoft. Best for tabular data (spreadsheet-like data with rows and columns).	Like asking 500 different experts a question, and taking a weighted vote. Each expert is simple, but together they're very accurate.	Our primary ML model. LightGBM regressor with 5,000 trees, learning rate 0.01, max depth 6. Research shows LightGBM > LSTM for stock prediction.
XGBoost	Another tree-based ML algorithm, similar to LightGBM but slightly slower. Made by University of Washington. Very popular in competitions.	Like LightGBM's older sibling -- equally smart but takes a bit longer to think.	Referenced in our research as an alternative. LightGBM is preferred because it's faster for our daily 50-stock scan.
Ensemble	Combining multiple ML models to get better predictions. Like getting a second opinion from multiple doctors.	Like asking 3 financial advisors and going with the majority recommendation. Each might miss something, but together they cover more ground.	v4's composite scorer IS an ensemble -- it combines ML score, VWAP, ORB, RS, FII, OI, and volume signals. Not just one model, but 7 complementary signals.
Classification vs Regression	Classification = predicting a category ("BUY or AVOID"). Regression = predicting a number ("stock will go up 2.3%").	Classification is like sorting mail into "urgent" and "not urgent" boxes. Regression is like predicting exactly how many minutes each letter will take to read.	v3 used classification (BUY/AVOID) -- this was a MISTAKE. A 5% gain and a 0.6% gain both became "BUY." v4 uses regression (predicts actual return magnitude) to preserve this information.

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Binary Classification	Classification with only 2 possible outcomes (yes/no, BUY/AVOID, 1/0). Simpler than multi-class but loses nuance.	Like a thumbs up / thumbs down review -- you lose the nuance of a 1-5 star rating.	What v3 did: `label = 1 if 5-day return > 0.5%, else 0`. This destroyed magnitude info and caused 96% AVOID signals. v4 moved away from this.
Target Variable	What the ML model is trying to predict. The "answer key" that the model learns from.	Like the answer column in a practice test. The model sees the questions (features) and tries to match the answers (target).	v4 target: intraday return = (close at 3PM - open at 9:30AM) / open. This tells the model "learn to predict how much a stock will move today."
Feature	A piece of information (input variable) that the model uses to make predictions. Like "RSI", "volume ratio", "VWAP deviation" -- each is a feature.	Like the ingredients in a recipe. Each ingredient contributes to the final dish. Features are the ingredients that produce the prediction.	v4 has 19 features: 9 daily context + 5 intraday + 3 institutional + 2 relative strength. Listed in `config.py`.
Feature Engineering	The art of creating useful features from raw data. Raw data: "price was Rs 100, 101, 103, 102." Engineered feature: "stock is in an uptrend with 0.8% daily momentum."	Like a chef turning raw ingredients into a prepared dish. Raw onions are not useful in a recipe, but caramelized onions add flavour. Feature engineering transforms raw data into something the model can use.	What `features_intraday.py` and `features_institutional.py` do. They transform raw price/volume/options data into meaningful features like VWAP deviation, ORB signal, and OI buildup.
Feature Importance	How much each feature contributes to the model's predictions. Higher importance = that feature is more useful.	Like finding out which ingredient makes or breaks a dish. If removing salt ruins everything but removing oregano makes no difference, salt has higher "feature importance."	v3 diagnosis showed ATR was 13.25% importance, MACD 7.56%, pct_from_low 6.21%. This revealed v3 was mean-reversion-heavy and missing momentum features.

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SHAP Values	A method to explain WHY the model made a specific prediction. For each stock, SHAP tells you which features pushed the score up or down.	Like a judge explaining their verdict -- "I gave you a high score because of strong VWAP position (+5 points), good ORB breakout (+3 points), but penalized for low volume (-2 points)."	Used in our ML research to identify the 20 most predictive features (Tier 1, 2, 3). SHAP revealed RSI and VWAP deviation as top features.
Permutation Importance	Another way to measure feature importance. Shuffle one feature's values randomly and see how much worse the model gets. Bigger drop = more important feature.	Like removing one player from a cricket team and seeing how much worse the team performs. If removing Virat Kohli causes a 50% drop in performance, his importance is very high.	Used alongside SHAP for feature selection. Helps decide which of the 19 features to keep.
Training Data	Historical data the model learns from. Like textbooks for a student. TradePilot uses 1-3 years of past Nifty 50 daily data as training data.	The practice exams a student studies before the real exam. More and diverse practice data = better prepared student.	1-year rolling window of daily data for all 50 Nifty stocks. Model trains on 12,000+ data points (50 stocks x 250 trading days).
Test Data	Data the model has NEVER seen, used to check if it actually learned useful patterns. If it works on test data, it will likely work on real trades.	The actual exam (not the practice one). If you only did well on practice but failed the real exam, you memorized instead of learning.	1-month forward window. Walk-forward validation uses each month as a test set while training on the preceding year.
Overfitting	When the model memorizes training data patterns (including noise)	Like a student who memorizes exact exam answers instead of understanding concepts. They ace the	v3's 44.79% precision on training data suggested it wasn't even fitting well. But overfitting is why we use walk-forward validation -- to catch it early.

Term	What It Is	Real-World Analogy	How TradePilot Uses It
	instead of learning general rules. It scores perfectly on old data but fails on new data.	old papers but fail when questions are rephrased.	
Underfitting	When the model is too simple to capture the real patterns. It performs poorly on both training AND test data.	Like a student who barely studied -- they can't even answer the practice questions, let alone the exam.	v3 was actually underfitting (poor training precision) because it used the wrong features and wrong target variable. v4 fixes this.
Walk-Forward Validation	The CORRECT way to test ML models on time-series data. Train on the past, test on the future, slide the window forward. Never let the model "peek" at future data.	Like predicting tomorrow's weather using ONLY data from today and before. You can't use Thursday's weather to predict Wednesday's. Time moves forward.	Mandatory in v4. Train on 1 year, skip 5-day embargo, test on 1 month. Slide forward and repeat. This is why k-fold is wrong for stocks.
K-Fold Cross-Validation	A standard ML testing method where data is randomly split into K parts (folds). The model trains on K-1 parts and tests on 1 part, rotating through all K parts.	Like shuffling exam papers and randomly picking some for practice and some for testing.	THIS IS WRONG FOR STOCKS because random shuffling mixes future data with past data. A 2025 data point could end up in training while 2024 is in testing -- that's time travel! v4 uses walk-forward instead.
Purged Cross-Validation	A modified version of k-fold that respects time order AND removes data near the train/test boundary to prevent information leakage.	Like leaving a gap between the practice exam and the real exam to make sure no questions from the real exam leaked into practice.	What we should use if we ever need cross-validation. The "purge" removes overlapping data, and the "embargo" adds a time gap.

Term	What It Is	Real-World Analogy	How TradePilot Uses It
Embargo Period	A gap (typically 5 days) between training data end and test data start. Prevents the model from learning from data that's "too close" to the test period.	Like a quarantine period between exposure and testing -- making sure no "contaminated" information leaks through.	v4 uses a 5-day embargo: `[===train 1yr===]--5d embargo--[test 1mo]`. This prevents the model from learning patterns that only work because of temporal proximity.
Information Coefficient (IC)	The correlation between what the model predicted and what actually happened. IC > 0.05 consistently = the model has a tradeable edge.	Like checking if a weather forecaster is actually accurate. If they say "70% chance of rain" and it rains 70% of the time, their IC is high.	THE key metric for v4. IC > 0.05 = usable. IC > 0.10 = strong. IC positive in > 60% of walk-forward folds = stable model.
R-squared	How much of the actual price movement the model explains. R-squared of 0.02 means the model explains 2% of price variance. Sounds low, but even 2% is profitable in trading!	Like how much of your exam score is explained by hours studied. Even if studying explains only 20% (other factors: mood, difficulty, luck), it's still worth studying.	LightGBM research showed R-squared of 2.13% monthly (2x better than simple linear regression). In trading, even small edges compound into big returns.
MAE (Mean Absolute Error)	The average size of prediction errors, ignoring direction. If the model predicts +2% but actual was +3%, the error is 1%. Average this across all predictions.	Like measuring how far off your estimated travel time is from actual. If you're usually off by 5 minutes, your MAE is 5 minutes.	Used as the loss metric for LightGBM training: `'metric': 'mae'`. Lower MAE = more accurate predictions.
Early Stopping	Stop training when the model starts getting worse on	Like a student who has studied enough and should stop. More studying after this	v4 uses 5,000 potential trees with early stopping. The model might stop at 2,000 trees if that's where performance peaked.

Term	What It Is	Real-World Analogy	How TradePilot Uses It
	validation data, even if you planned more training rounds. Prevents overfitting.	point actually hurts because they start memorizing instead of understanding.	
Hyperparameters	Settings that control HOW the model learns (not WHAT it learns). Like the volume, bass, and treble knobs on a speaker.	Like adjusting a recipe -- how much salt (learning rate), how long to cook (n_estimators), how big the pot (max_depth). You tune these before cooking.	v4 hyperparameters: `num_leaves=31, max_depth=6, learning_rate=0.01, min_child_samples=100, subsample=0.7`. Each was chosen based on ML research.
Learning Rate	How fast the model updates its knowledge with each new tree. Too high = overshoots. Too low = learns too slowly.	Like the speed of a car approaching a parking spot. Too fast and you overshoot. Too slow and you never get there. 0.01 is a careful approach.	Set to 0.01 (very conservative) with 5,000 trees. The model takes tiny steps and uses many iterations, which produces more robust predictions.
Winsorization	Capping extreme values to reduce the impact of outliers. If a stock gained 15% in a day (rare), cap it at the 99th percentile (maybe 5%).	Like grading on a curve -- if one student scored 100% and everyone else scored 40-60%, you might cap the 100% at 80% so it doesn't distort the average.	Applied to target variable: `np.clip(target, target.quantile(0.01), target.quantile(0.99))`. Extreme one-day moves (circuit limits, news shocks) are capped to prevent the model from being distorted by rare events.

Section 6: Portfolio & Position Sizing

How TradePilot decides how MUCH money to put into each stock.

Term	What It Is	Real-World Analogy	How TradePilot Uses It
Kelly Criterion	A mathematical formula that calculates the optimal bet size based on your edge (win rate and payoff ratio).	Like a professional poker player who calculates exactly how much to bet based on their odds.	Implemented in `position_sizer.py`. Formula: `f = p - q/b` where p = win rate, q = loss rate, b = payoff ratio.

Term	What It Is	Real-World Analogy	How TradePilot Uses It
	Invented by a Bell Labs scientist for information theory.	They never go "all in" on a mediocre hand.	Used as a reference for maximum position sizing.
Half-Kelly	Using HALF of what Kelly Criterion suggests. Full Kelly is mathematically optimal but emotionally brutal (high volatility). Half-Kelly is smoother.	Like driving at 60% of the speed limit instead of 100%. You arrive slightly slower but MUCH safer.	v4 uses Half-Kelly: `half_kelly = full_kelly / 2.0`, capped at 25% of capital. With 55% win rate and 2:1 payoff, Half-Kelly = ~5% per trade.
Position Sizing	Deciding how many rupees (and shares) to invest in each stock. Higher-confidence signals get larger positions.	Like allocating your monthly budget -- rent gets the most (high priority), entertainment gets less. You spend more on things you're more confident about.	Score-weighted allocation: `alloc_i = (score_i / total_scores) * capital`. Higher composite scores get proportionally larger positions. Max 20% in any single stock.
Score-Weighted Allocation	Distributing capital proportionally based on composite scores. A stock scoring 74 gets more capital than one scoring 64.	Like a teacher distributing prizes based on marks. The top scorer gets the biggest prize, second gets slightly less, and so on.	Core allocation logic: if 10 BUY stocks have total score of 700, a stock with score 74 gets 74/700 = 10.6% of capital.
Risk Per Trade	The maximum amount you're willing to lose on a single trade. Usually 1-2% of total capital.	Like the maximum you'd bet in a poker hand. Even with good cards, you never risk more than you can afford to lose.	Calculated as `risk_rs = qty * price * (sl_pct / 100)`. If you buy 100 shares at Rs 1,000 with 1% SL, your risk is Rs 1,000.
Maximum Drawdown	The largest peak-to-trough decline in your portfolio value. If your portfolio went from Rs 10L to Rs 8L at worst, max drawdown is 20%.	Like the worst dip in a roller coaster. Your stomach drops the most at the biggest fall. Max drawdown measures the scariest moment in your investment journey.	Tracked in portfolio risk metrics. The VWAP strategy in our research had only 16% max drawdown (very good). Pairs trading had 2.57% (excellent).
Sharpe Ratio	A measure of risk-adjusted returns. Higher = better. It asks: "How much return am I getting for each unit of risk I'm taking?" Sharpe > 1 = good, > 2 = excellent.	Like measuring a restaurant's value: not just "is the food good?" but "is the food good FOR THE PRICE?" A Rs 200 thali that's 8/10 quality has a better	Used to evaluate strategies. ORB strategy: Sharpe 2.396 (excellent). VWAP strategy: Sharpe 3.57 (outstanding). Our target is Sharpe > 1.5.

Term	What It Is	Real-World Analogy	How TradePilot Uses It
		Sharpe than a Rs 2,000 dinner that's 9/10.	
Profit Factor	Total profits divided by total losses. If you made Rs 50,000 in winning trades and lost Rs 25,000 in losing trades, profit factor = 2.0. Above 1.5 = good.	Like a business's revenue-to-cost ratio. If you earn Rs 2 for every Rs 1 you spend, your profit factor is 2.0.	Tracked across paper trading results. Our target is profit factor > 1.5.
Win Rate (Hit Rate)	The percentage of trades that made money. 55% win rate = 55 out of 100 trades were profitable.	Like a cricket batsman's strike rate -- not every ball is a boundary, but you need to score often enough to win.	With 55% win rate and good risk-reward (1:2), you're profitable even though you lose 45% of trades. Our target is >53% win rate. v4 combines high win rate signals (ORB at 60-89%) with good sizing.
Capital Allocation	How you split your total capital across different stocks and cash. Not putting all eggs in one basket.	Like dividing your savings: 60% in stocks, 20% in fixed deposits, 20% in cash. Within stocks, further splitting across sectors.	v4 sizes positions so no single stock gets >20% of capital (`max_per_stock_pct = 0.20`). Minimum Rs 20,000 per position. Rest stays as cash (10-30% cash buffer).
Diversification	Spreading investments across multiple stocks/sectors to reduce risk. If one stock crashes, others may hold up.	Like not putting all your eggs in one basket. If you carry 3 baskets and drop one, you only lose 1/3 of your eggs.	v4 recommends 5-10 positions per day. Position sizer automatically distributes across BUY signals to avoid concentration.

Section 7: Trading Strategies

The proven strategies that TradePilot's v4 engine draws from.

Term	What It Is	Real TradePilot Connection
Momentum Trading	Buying stocks that have been going UP, betting they'll continue going up. "Trend is your friend." Stocks in motion tend to stay in motion.	v4's relative strength ranking IS momentum trading. We buy the top performers among 50 stocks, betting they'll keep outperforming. The NSE Momentum 50 Index consistently beats Nifty 50.

Term	What It Is	Real TradePilot Connection
Mean Reversion	Buying stocks that have fallen "too much," betting they'll bounce back to their average. The opposite of momentum.	v3 was accidentally a mean-reversion system (ATR + pct_from_low were top features). v4 uses RSI and Bollinger %B for mean-reversion, but combines them with momentum signals.
Trend Following	Identifying the direction of the market (up/down/sideways) and trading in that direction. Don't fight the trend.	v4's market regime detection (BULL/BEAR/NEUTRAL) IS trend following. In a bull market, we're more aggressive with buys. In bear, we're cautious. MACD and ADX help identify trends.
Pairs Trading (Statistical Arbitrage)	Finding two stocks that move together (like HDFC Bank and ICICI Bank) and trading the spread between them. When one diverges, bet on convergence.	Referenced in our research. Nifty 50 pairs trading showed 4.97% annual return with only 2.57% max drawdown. Market-neutral strategy (works in bull and bear markets). Not implemented in v4 yet.
Cointegration	A statistical property where two stocks are "tied" together long-term even if they diverge short-term. Required for pairs trading.	Like two dogs on a leash -- they can wander apart, but the leash always pulls them back together. HDFC Bank and ICICI Bank are cointegrated.
Factor Investing	Ranking stocks by specific "factors" (momentum, value, quality, volatility) and buying the top-ranked ones. Used by Smallcase in India.	Like hiring employees based on specific traits (experience, education, references) rather than gut feeling. Each factor is a measurable trait.
Opening Range Breakout Strategy	Trade the breakout of the first 15-minute high/low. If price breaks above the high, buy. If below the low, sell/avoid.	Our #2 priority signal (15% weight). Research shows 60-89% win rate on NSE with Sharpe 2.396. Computed in `compute_orb()`. Filter with VWAP confirmation and ATR-based stop-loss.
VWAP Strategy	Buy when price is above VWAP (institutions are buying). Sell/avoid when below VWAP. Close all positions by end of day.	Research shows Sharpe 3.57 and 501% return over 5 years. v4 uses VWAP position as a 10% weighted signal. Deviation from VWAP is also an ML feature.
Sector Rotation	Moving money from weak sectors to strong sectors as the economic cycle changes. IT does well in some phases, banking in others.	Like a farmer rotating crops -- different crops do well in different seasons. Similarly, different sectors lead in different market phases.

Section 8: TradePilot-Specific Terms

Terms unique to our platform that you'll see in the codebase and dashboard.

Term	What It Is	Where In Code	Example
Composite Score	The final 0-100 score for each stock, combining all 7 signals. This is the "verdict" -- higher score = stronger buy signal.	`composite_scorer.py` -- weighted sum of ml_score, rs_score, orb_score, vwap_score, fii_score, oi_score, vol_score. Scaled to 0-100.	TITAN composite score: 74.2 (BUY). INDUSINDBK: 38.5 (AVOID).
Composite Weights	How much each signal contributes to the final score. Total = 100%.	`config.py` : ML=25%, RS=20%, ORB=15%, VWAP=10%, FII=10%, OI=10%, Volume=10%.	If ORB is strong (+0.9) but FII is negative (-0.5), ORB contributes 0.15*0.9 = 0.135 and FII contributes 0.10*(-0.5) = -0.05 to the score.
v2/v3/v4 Engine	TradePilot's algorithm versions. v2 = basic scoring with fixed thresholds. v3 = ML-based with binary classification (broken). v4 = composite scoring with regression (current).	v2 in `app.py` , v3 in `trading_engine_v3.py` , v4 in `prototype/v4/` .	v2 needed score >= 55 for BUY. v3 needed 50-60 (regime-dependent). v4 uses percentile ranking (top 20% = BUY).
Score Distribution	How scores are spread across all 50 stocks. A good distribution has scores spread from 30-80. A broken one (v3) had 96% of stocks scoring below 20.	v3 had range 0.5-69 with mean 17, median 12.6. v4 targets a more uniform distribution by using percentile ranking.	v3 problem: only 2 of 50 stocks scored above BUY threshold (55+). v4 fix: top 20% are ALWAYS BUY regardless of absolute score.
BUY / HOLD / AVOID	The three classification buckets for stocks. BUY = take a position. HOLD = already holding? keep it. AVOID = don't touch.	`config.py` : BUY = top 20% by percentile (always 10 stocks). HOLD = next 30% (15 stocks). AVOID = bottom 50% (25 stocks).	With 50 stocks: always 10 BUY, 15 HOLD, 25 AVOID. This fixes v3's problem of having only 2 BUY signals.
Relative Ranking (Percentile)	Ranking each stock's score as a percentile among all 50 stocks. Rank 1/50 = percentile ~1.0. Rank 25/50 = percentile ~0.5.	`composite_scorer.py` : `rank = np.searchsorted(sorted_rs, rs_today) / len(sorted_rs)` .	If TITAN is ranked #2 out of 50, its percentile = 0.96. If NESTLEIND is #40/50, its percentile = 0.20.
Signal	Any piece of evidence pointing toward BUY	The `reasons` list in each stock's output. Each reason has a `type`	TITAN signals: "ORB breakout above 3210"

Term	What It Is	Where In Code	Example
	or AVOID. A stock might have 5 positive signals and 2 negative signals.	(positive, negative, neutral) and `text`.	(positive), "Price +0.40% above VWAP" (positive), "FII buying (+1500 Cr)" (positive), "Volume surge (2.1x)" (positive).
Consensus Signal	When BOTH the v3 engine and v4 engine agree on a BUY or AVOID signal. Double confirmation = higher confidence.	When v4 is fully operational, the dashboard can show consensus between engines. Currently only v4 runs.	If v3 says BUY and v4 says BUY for TITAN, it's a consensus BUY -- very high confidence.
Market-Wide Signal	Signals that apply to ALL stocks equally (not stock-specific). FII/DII flow, Nifty change, India VIX, advance-decline ratio.	`nifty_change_pct`, `fii_net_crores`, `dii_net_crores`, `india_vix`, `advance_decline_ratio` in config.py.	FII buying +1,500 Cr is a market-wide positive signal -- it boosts ALL 50 stocks' FII sub-score equally.
Per-Stock Signal	Signals specific to one stock -- its VWAP position, ORB breakout, options OI, relative strength. Different for each stock.	`vwap_deviation_pct`, `orb_signal`, `oi_buildup_signal`, `rs_vs_nifty_5d` in config.py.	TITAN's ORB breakout is a per-stock signal. RELIANCE might still be inside its ORB range (no breakout).
Scan Cycle	One complete run of the v4 engine -- fetching data for all 50 stocks, computing all features, scoring, ranking, and generating signals. Takes 30-60 seconds.	`score_all_stocks()` in `composite_scorer.py`. Fetches market data, batch quotes, per-stock intraday candles, options chains.	A typical scan cycle: "Scoring 50 stocks... Got quotes for 48... BUY=10 HOLD=15 AVOID=23... 42.3s elapsed."
Force Exit	Mandatory closure of all positions at a specific time (3:15 PM for intraday). No exceptions -- even if a stock is mid-rally, you exit.	Not yet coded in v4, but referenced in VWAP strategy research: "Close all EOD." Essential for intraday risk management.	At 3:15 PM, if SBIN is up 1.8% and still rising, you still sell. Overnight risk is not worth the potential extra gain.
Paper Trading Pool	A simulated portfolio with Rs 10,00,000 that tracks all BUY signals	`position_sizer.py`: `capital = 1_000_000.0`. Positions are sized, tracked, and P&L is calculated as if real.	Today's paper portfolio: 10 positions, Rs 9.2L deployed, Rs

Term	What It Is	Where In Code	Example
	as if we had actually traded them. Measures real-world performance without real money.		80K cash. Expected P&L: +Rs 5,200 (if 55% win rate holds).

Quick Reference Card

The v4 Composite Score Formula

```
Composite Score = 0.25 x ML Prediction
                + 0.20 x Relative Strength Rank
                + 0.15 x ORB Breakout Score
                + 0.10 x VWAP Position Score
                + 0.10 x FII/DII Flow Score
                + 0.10 x Options OI Score
                + 0.10 x Volume Score
```

What Changed from v3 to v4

Aspect	v3 (Broken)	v4 (Fixed)
Prediction type	Binary classification (UP/DOWN)	Regression (predict actual % return)
Features	10 mean-reversion features	19 features across 4 categories
BUY signal method	Fixed threshold (score > 55)	Percentile ranking (top 20%)
Typical BUY count	2 out of 50 (96% AVOID)	10 out of 50 (always)
Intraday signals	None	VWAP, ORB, gap analysis, momentum
Institutional data	None	FII/DII flow, options OI, PCR
Position sizing	Equal weight	Score-weighted with Kelly reference

Numbers to Remember

Metric	Good	Great	TradePilot Target
Win Rate	>53%	>60%	55-60%
Sharpe Ratio	>1.0	>2.0	>1.5
Profit Factor	>1.2	>2.0	>1.5

Metric	Good	Great	TradePilot Target
Max Drawdown	<20%	<10%	<15%
Information Coefficient	>0.05	>0.10	>0.05 consistently
Risk-Reward Ratio	>1.0	>2.0	>1.5

Last updated: April 8, 2026 Source files: algorithm_diagnosis.md, ml_training_pipeline.md, opensource_trading_algos.md, config.py, composite_scorer.py, features_intraday.py, features_institutional.py, position_sizer.py